

week4

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What I did

- Completed with the data cleaning and modelling tasks.
- Finished the definition of cross validation and evaluation of models.
- A small experiment: Compare the results of leaving structural missing values as `np.nan` against filling them with an explicit string category. And also just remove these columns.

Experiment method

String Version

```
model_df_str = model_df.copy()
```

```
model_df_str[exp_cols] = model_df_str[exp_cols].fillna("N/A")
```

#NaN Version

```
model_df_nan = model_df.copy()
```

#Drop version

```
model_df_drop = model_df.dropna(subset=exp_cols).copy()
```

The way dealing with miss values in Patient Health Questionnaire-4 and Pre-existing health conditions

```
split_and_save(processed_nan, "exp_nan")  
split_and_save(processed_str, "exp_str")  
split_and_save(processed_drop, "exp_drop")
```

Gridsearch settings

```
grid_xgb = {  
    'learning_rate': [0.05, 0.1, 0.2],  
    'n_estimators': [100, 250, 500],  
    'max_depth': [4, 6, 8],  
    'colsample_bytree': [0.6, 0.8, 1.0]  
}  
xgb_clf = XGBClassifier(use_label_encoder=False, eval_metric=
```

- Based on the results of the preliminary experiment, employing explicit string imputation did not compromise the model's predictive performance.
- Removing these specific features contributes valuable signals to the overall predictions.
- Because most models to be evaluated cannot directly handle np.nan values, I will proceed with the explicit string imputation strategy for the remainder of the analysis. ## Results NaN version ROC AUC: 0.898 (0.002) Str version ROC AUC: 0.898 (0.002) Drop version ROC AUC: 0.899 (0.002)

Plan for next week

- Model tuning.
- Begin my presentation slides
- Begin my final report

Model tuning (w3-6)

Fit four models: Logistic Regression, Classification Trees, XGBoost, and MLPs. For the tree based models, hyperparameters will be tuned with 5-fold cross validation on the training data, with AUC as the optimisation metric.

Model comparison (w6-7)

Compare the four models using AUC, Precision, Recall, Accuracy, and F1 score. The best models will be selected based on predictive performance.

Interpretation (w7-9/10)

Investigate the stable important features by SHAP values on the two best performing models.